

3 Properties of the Lagrangian

$$\begin{aligned}\mathcal{L} &= \mathcal{L} + c \\ \mathcal{L} &= c\mathcal{L}\end{aligned}$$

In effect, because when we use the Euler Lagrange equation, nothing changes.

because there are derivatives, so the changes wouldn't have any effect.

$$\frac{d}{dt} \frac{d\mathcal{L}}{d\dot{x}} = \frac{d\mathcal{L}}{dx}$$

If we consider a free particle with $\vec{F} = 0$, $\mathcal{L} = T + \text{constant potential} = T$

Bead on a rotating wire